Technical Report - Math Wizard 2.0

1. Cover Page

- **App Name**: Math Wizard
- **Team**: Arlindo Dishi
- **Date**: April 2026

2. Introduction

Problem Statement: Young students often struggle to engage with simple arithmetic problems, and default applications often lack strong accessibility options. **Solution**: Math Wizard 2.0 is an interactive, gamified Android application that quizzes students on arithmetic. It utilizes auditory (Text-to-Speech) and tactile (Haptic Vibration) feedback to make learning highly accessible. We also implement a dedicated High Contrast visual mode for users with visual impairments.

3. Architecture Overview

The application uses the **MVVM (Model-View-ViewModel)** architectural pattern natively in Kotlin.

- `MainActivity` serves as the routing hub.
- `DifficultyActivity` routes the selected level of challenge.
- `QuizActivity` relies on `QuizViewModel` for all business logic (generating random equations, validating answers, applying logic).
- `SettingsActivity` controls the deep hardware/software toggles via `SharedPreferences`.
- `HighScoreActivity` (Statistics) tracks the mathematical accuracy of the player over time.

4. Feature Description

1. **Multi-screen App**: Handled cleanly with Explicit Intents (`Intent(this, Target::class.java)`).
2. **UI Layouts**: Fully layered `ConstraintLayout` screens utilizing modern `MaterialCardView` and `MaterialButton` components.
3. **Data Persistence**: `SharedPreferences` saves Total Played, Total Correct, High Score, High Contrast Mode, and Haptic/TTS Toggles.
4. **Hardware Access - Text-To-Speech (TTS) & Haptics**: Uses `TextToSpeech` engine and the `Vibrator` device architecture to create a fully immersive physical mapping. Correct answers issue short, light vibrations, while wrong answers issue heavy, multi-wave vibrations.

5. UI/UX Design Decisions

- **Material Cards**: Adding shadowing to the Question Block gives the UI a premium floating aesthetic.
- **Accessibility - High Contrast**: Instead of simply making the app 'dark', High Contrast mode forcibly recolors the layout backgrounds to true black (`#000000`) and the text dynamically to true yellow (`#FFFF00`). This is an essential accessibility pattern.

6. Testing

#	Test Case Description	Device / API Level	Status
1	Navigate MainActivity -> Difficulty -> Quiz	Emulator API 34	PASS
2	High Contrast mode accurately flips UI colors	Emulator API 34	PASS

3	Haptic feedback triggers specifically on button tap	Physical Device API 34	PASS
4	Statistics accurately calculates percentage score	Emulator API 34	PASS

7. Reflection

- **What went well:** Pushing beyond standard visual interfaces by implementing Vibrator APIs effectively anchors the learning experience.
- **What could be improved:** Future iterations could persist all incorrect questions inside a local SQLite Room database, so the child can revisit exactly which questions they frequently miss.